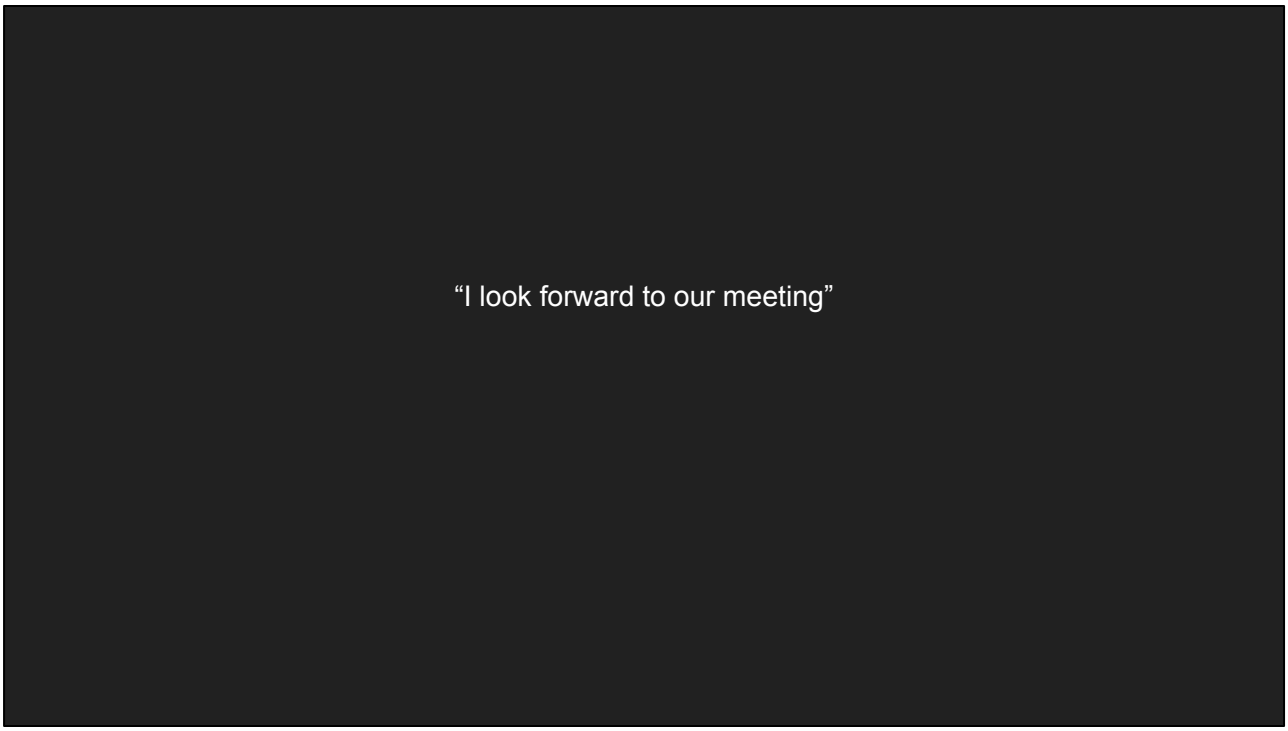
Time is Made Up

A Linguistic Perspective

Jason Don #linguistic-tools

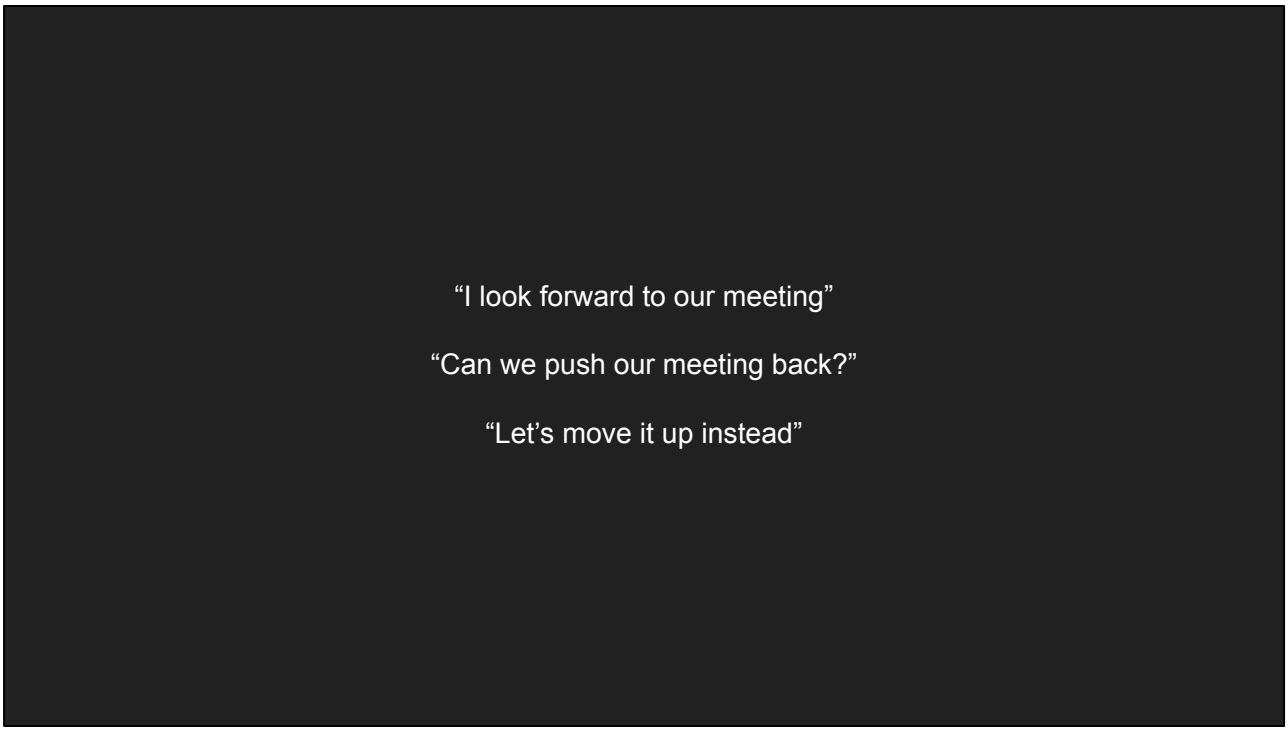
Hello! I'm Jason and I'm going to talk about time from a linguistic perspective.
The idea for this talk came when as usual, the linguistic tools team went down a rabbit hole during an important meeting



The rabbit hole starts here. We often say things like I look forward to something in the future, implying that the event is physically in front of us. Time is a completely abstract concept and we can’t point in a direction and say oh that’s where the future is. But in english, we give time a spatial dimension to make it easier to understand. So the future is in front of us, the past is behind us, that makes sense right?



But then we say things like “can we push the meeting back” and we’re asking for the meeting to be moved further into the future. Push it… back… to tHe fUtuRE? Actually, we can explain this. If there’s an object in front of us we usually want IT to back up so we push it backwards. So time is still a dimension, future events are objects in front of us, and we push them back away from us further into the future. Okay so what do we say when we want to move a meeting earlier?



we move time… up? This is unfortunately consistent with our object metaphor. When we describe something as close to us we say up close like up close and personal, bring it up to me. Directionally we’re all over the place but technically, we are consistent.
So this is a part of linguistics, the scientific study of language. For us on linguistic tools, we apply this to computer science to build tools like spelling correction and machine translation. Unfortunately, natural language processing has been moving away from classical linguistics and towards statistical machine learning for quite some time now

Anytime a linguist leaves the group the recognition rate goes up

- Frederick Jelinek

A speech recognition researcher is often quoted with saying, “anytime a linguist leaves the group the recognition rate goes up”. I prefer a quote from my favorite linguist

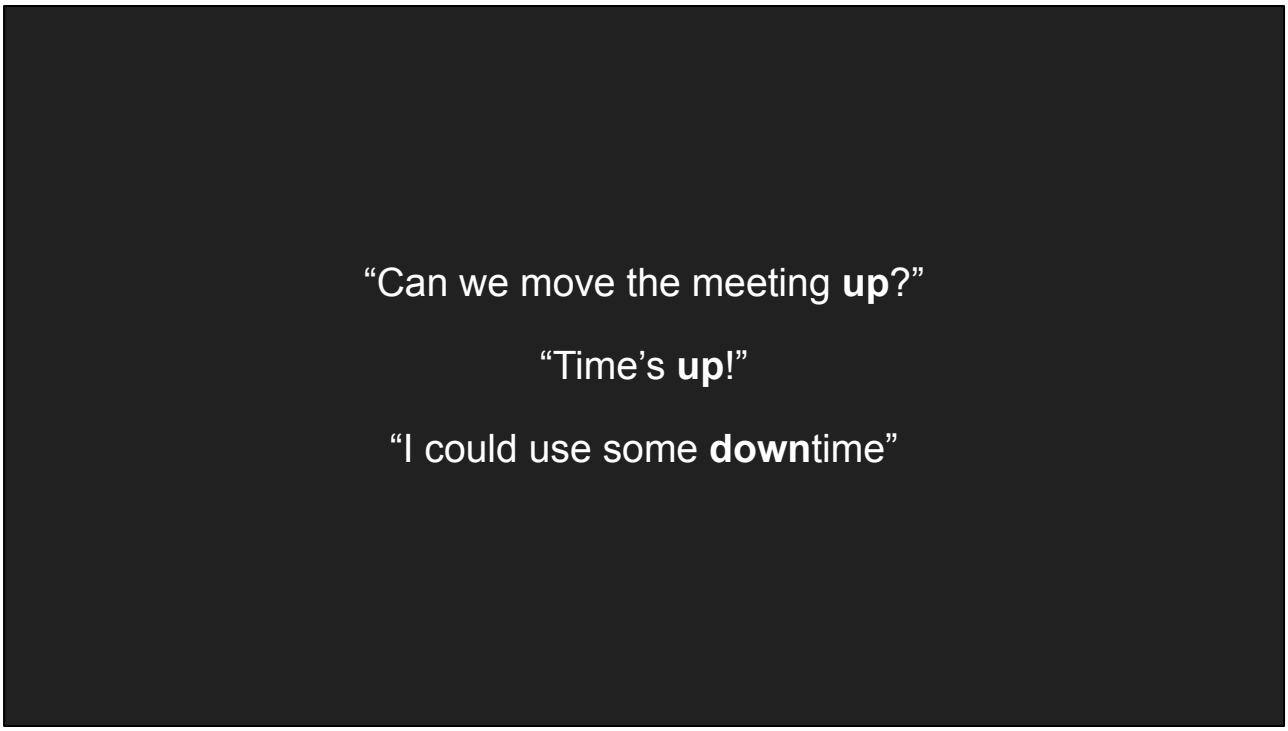


Why waste time say lot word when few word do trick?

Okay, but back to time...

Time is a dimension

So like we've discussed, time is a dimension and in English, we see it as a one dimensional line. Sometimes we use it as forwards and backwards, other times we use timelines that go from left to right.



We also use “up” like moving a meeting up or “time’s up”! While there is no moving meetings down or time’s down, we do have an uptime vs downtime analogy



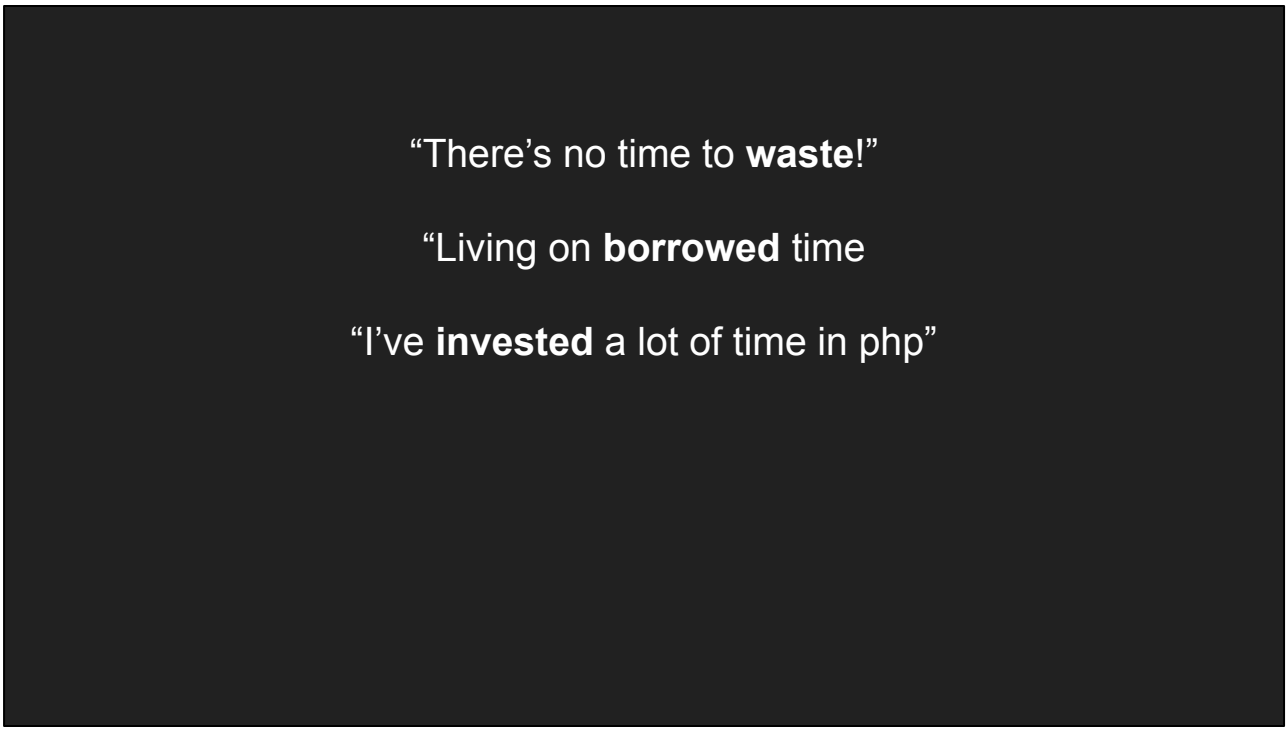
In mandarin time is also dimensional but moves in the opposite direction! When we say day after tomorrow hou tien or when we say in the future yi hou, we use the word for behind/rear. Then for the past, we say qian tien and yi qian which uses the word for forwards. This is not wrong, just a different perspective. You can think of time as a book where the beginning of time is at the front of the book while the end of time is at the back.

Time is money

Another common metaphor we use in english is the idea of time being money and the idea that time is a valuable commodity



Everyday we spend time, save time, budget our time, always thinking about the value of our time



In some darker ways we see time as something that can be wasted. Something that can be borrowed which implies that we will owe time to someone or something. Then worst of all, we invest time to suggest that we’re taking a risk and expect returns. I guess it’s a good things stocks only go up right?

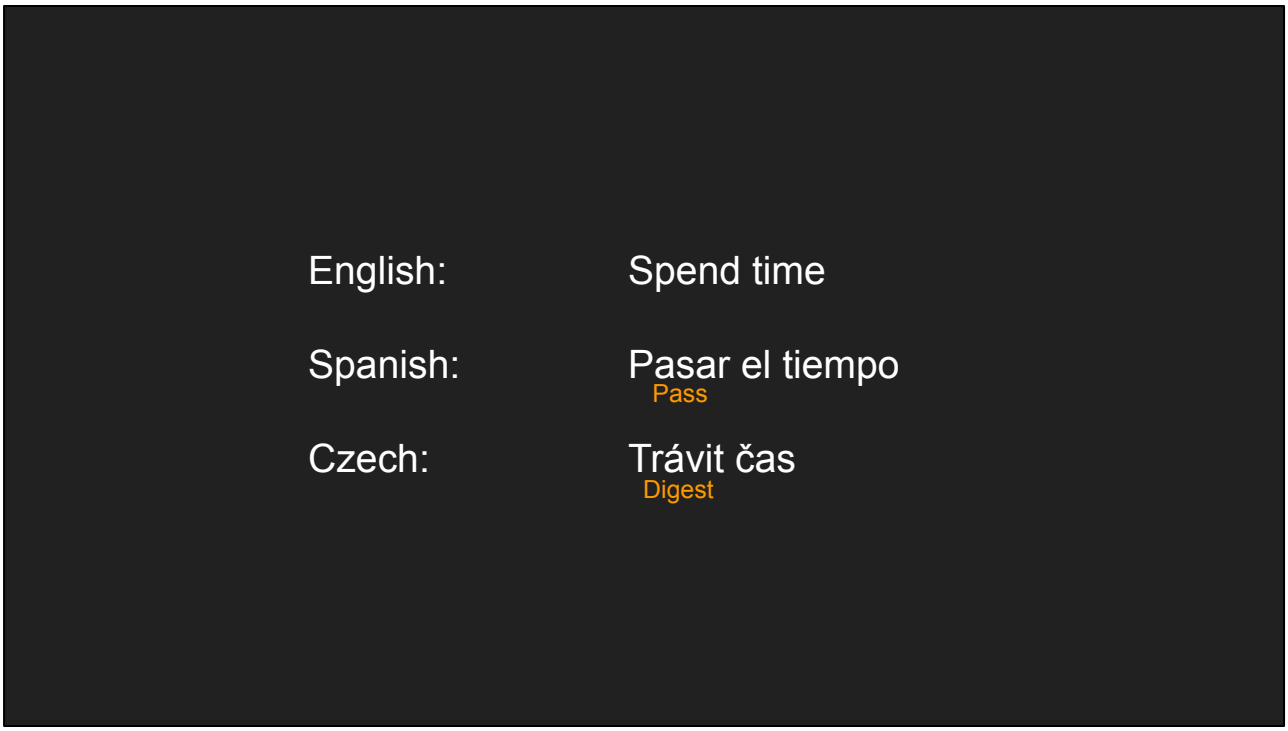
“There’s no time to **waste**!”

“Living on **borrowed** time

“I’ve **invested** a lot of time in php”

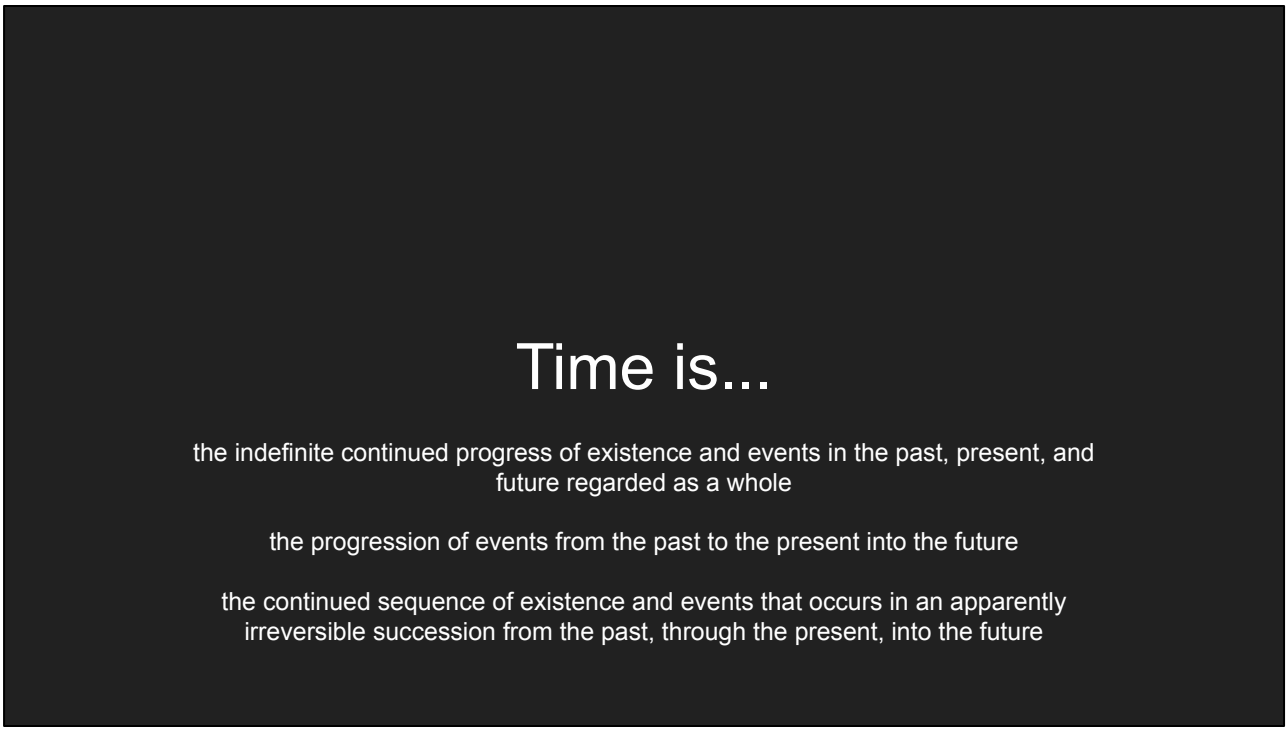


Can’t wait for the returns on my push queue wait times



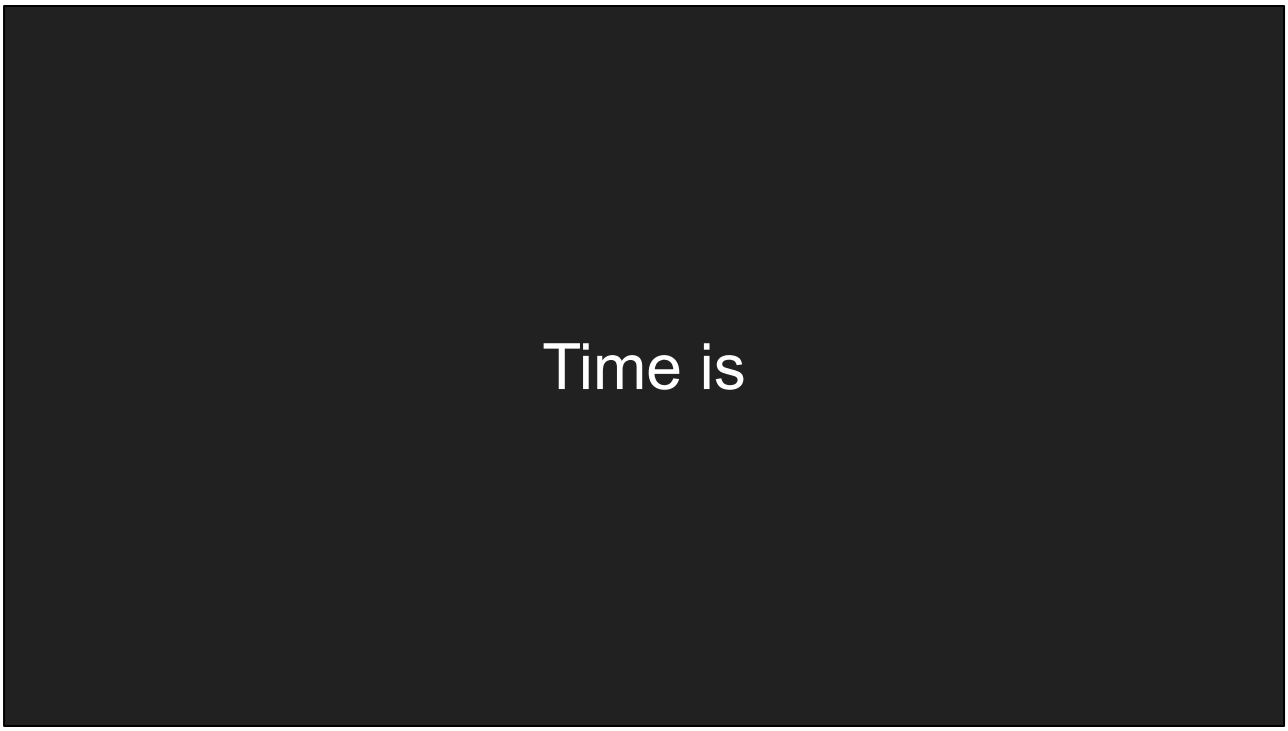
The phenomenon is not universal. In spanish you don't spend time but rather you pass time. In czech, you don't spend it, you digest it! I think it's funny to think of time as something that you're continually eating because it really takes enjoying your time to a whole new level.

It's not perfect, these languages have other ways of describing time as money, but it's fun to think about how we would view time from a different lens



So to bring it all together, what is time, really? We have some definitions, but they all use the concept of time in their definitions. Using the past present and future to define time is super circular. To make it easier to understand, we have these metaphors. Is time money? A dimension? A digestive tract? In physics, time is relative? relative to light? Relative to me and you? Maybe time is just the speed of the computer that's simulating our reality

On a more personal note, I can't help but feel the pressure to not waste time. The internet constantly tells us to pack more into our schedules and to not waste the time we have. I can't tell you what time really is but I want to acknowledge that our language mirrors our perspective. What if we didn't see time as a limited resource, but rather as the naturally occurring phenomena that it is?



We would never say we’re wasting gravity or that we’ve spent too much light ya know? So smell the roses, pet the dogs. Time is made up so we might as well enjoy it

Some sources + fun links that I didn't include

- translate.google.com
- https://en.wikipedia.org/wiki/Metaphors_We_Live_By
- https://memory.ai/timely-blog/linguistic-relativity-concept-of-time
- 6 cool facts about time in different cultures with anecdotes https://www.k-international.com/blog/time-in-different-languages-and-cultures/
- https://www.popsci.com/language-time-perception/
- Catalan's time system uses +1 indexing, swahili clocks are upside down, hopi time controversy? https://www.quora.com/Which-languages-tell-the-time-in-interesting-ways
- https://www.babbel.com/en/magazine/language-affect-perception-time
- http://language.ucsd.edu/papers/language-time.pdf
- Famous linguists: Robin Lakoff, Erving Goffman, Eve Sweetser, Edward Sapir, Benjamin Lee Whorf

Team channel:	#linguistic-tools
Fun language enthusiasts:	#language-interest

That's my time! If you're a language nerd, come find us in the linguistic-tools channel or the language interest channel, thanks everyone :D